Flood Detection and Post-Flood Recovery Assessment in Sylhet District, Bangladesh Using Sentinel-1 SAR and Night-Time Light Data

1. Abstract

Flooding is a recurring natural hazard in Bangladesh, significantly impacting socio-economic systems, particularly in low-lying regions such as Sylhet. This study analyzes the 2017 flood event in Sylhet District using Sentinel-1 Synthetic Aperture Radar (SAR) data to detect flood extent and VIIRS night-time light (NTL) data to assess post-flood recovery. A change detection approach using VV and VH polarization ratios was applied to extract flooded areas, followed by masking techniques to isolate flood-affected regions. The total flooded area was estimated at **377.83 km²**. Post-flood recovery was quantified using an NTL-based recovery index and analyzed in relation to population density, relative wealth index (RWI), and infrastructure proxy (road density) through linear regression. The regression results indicate that infrastructure positively influences recovery, whereas higher population density and wealth index show negative relationships with recovery. These findings highlight spatial disparities in resilience and emphasize the importance of infrastructure in post-disaster recovery.

2. Introduction

Bangladesh is one of the most flood-prone countries in the world due to its monsoon climate, dense river network, and low elevation. The northeastern region, particularly Sylhet, experiences frequent flash floods caused by heavy rainfall and upstream inflow from hilly regions. The 2017 flood event caused extensive damage to agriculture, infrastructure, and livelihoods.

Accurate flood mapping is essential for disaster management and planning. Optical satellite imagery is often limited during flood events due to cloud cover; therefore, Sentinel-1 SAR data provides a reliable alternative. Additionally, post-disaster recovery assessment is critical for understanding resilience. Night-time light (NTL) data has emerged as an effective proxy for economic activity and recovery analysis.

This study aims to:

- Detect flood extent using Sentinel-1 SAR data
- Extract flooded areas using masking techniques
- Assess post-flood recovery using NTL data
- Analyze the relationship between recovery and socio-economic factors

3. Methodology

3.1 Study Area

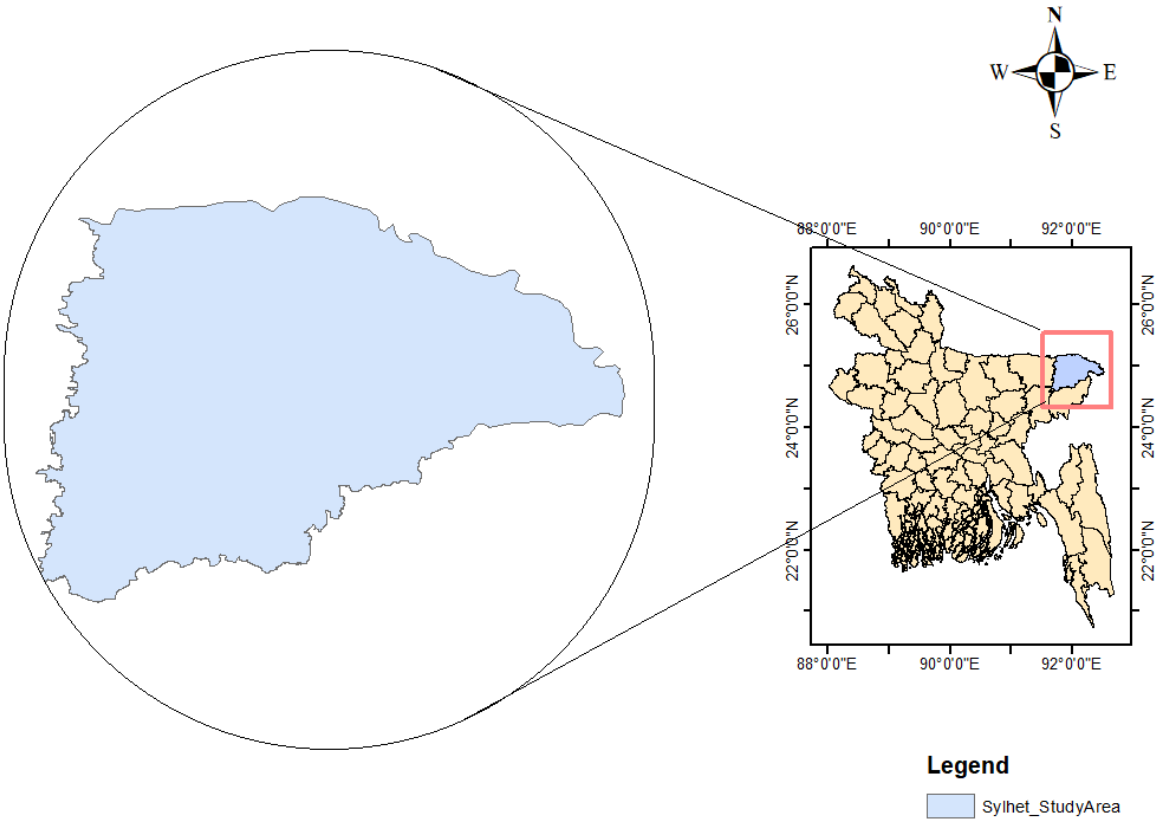

Figure 1 Study Area Map

The study was conducted in Sylhet District, located in northeastern Bangladesh, characterized by low-lying floodplains and haor ecosystems.

3.2 Data Sources

Data Type	Dataset	Purpose
SAR Data	Sentinel-1 (VV, VH)	Flood detection
Night Light	VIIRS VNP46A2	Recovery analysis
Population	WorldPop	Population density
Wealth	Relative Wealth Index (Facebook)	Economic condition

Infrastructure	GHSL Built-up Data	Road density proxy

3.3 Flood Detection Procedure

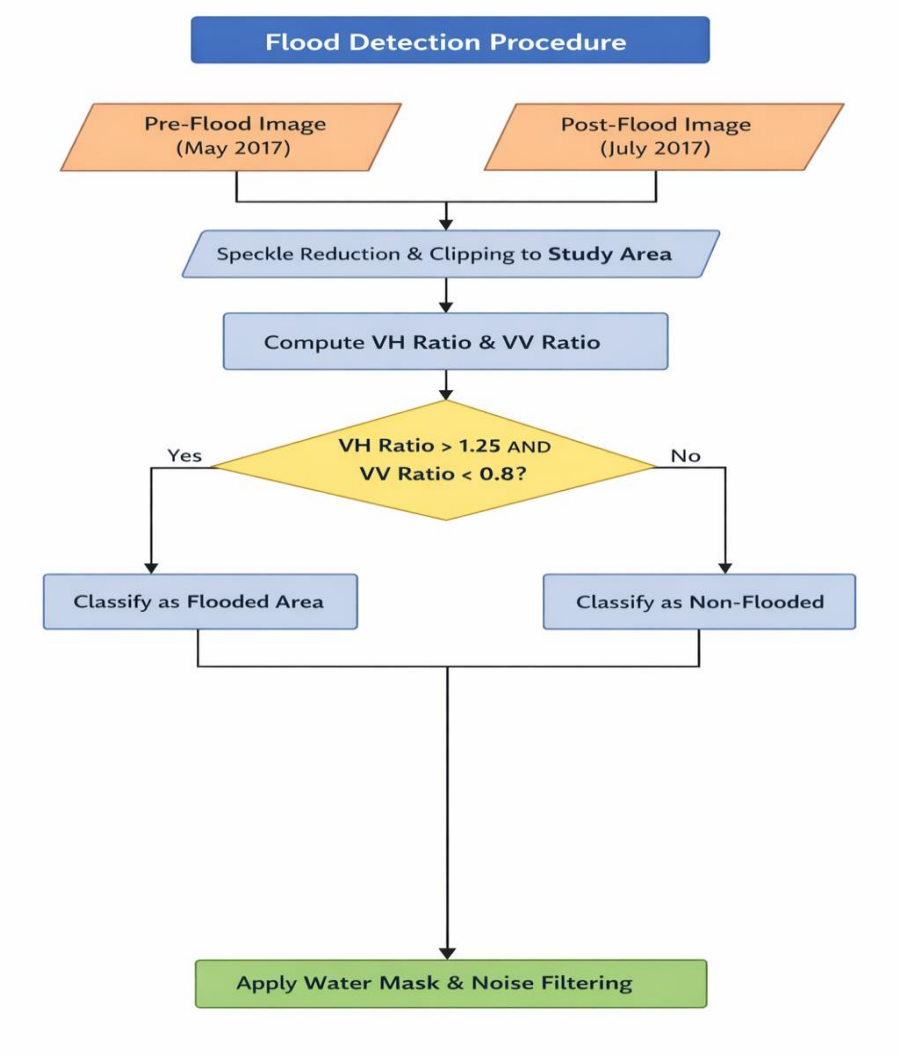

Figure 2 Procedure of flood detection

3.4 Flood Area Calculation

Flooded area was calculated using pixel area summation at 10 m resolution and converted into square kilometers.

3.5 Post-Flood Recovery Assessment

3.5.1 Night-Time Light Analysis

Three temporal phases were considered:

- Pre-flood (May 2017)

- Post-flood (October 2017)
- Recovery (Nov–Dec 2017)

3.5.2 Recovery Index Formula

To quantify recovery, a normalized recovery index was developed based on temporal changes in NTL intensity. The recovery index was calculated as:

$$\text{Recovery Index} = \frac{(\text{Recovery} - \text{Post})}{(\text{Pre} + 0.1)}$$

A small constant (0.1) was added to the denominator to avoid division by zero and to stabilize the index in areas with low baseline illumination. This index provides a relative measure of recovery, where higher values indicate greater restoration of nighttime activity following the flood event.

3.5.3 Socio-economic Variables

- Population density
- Relative Wealth Index (RWI)
- Road density (derived from GHSL data)

3.5.4 Regression Analysis

A multiple linear regression model was employed to assess the relationship between NTL-based recovery and the selected socio-economic variables. The model is expressed as:

$$\text{NTL Recovery} = \beta_0 + \beta_1(\text{PopDensity}) + \beta_2(\text{RWI}) + \beta_3(\text{RoadDensity})$$

where β_0 represents the intercept and β_1–β_3 represent the coefficients of the explanatory variables. The regression analysis was performed using 725 spatial samples extracted across the study area, enabling the identification of statistical relationships between recovery patterns and socio-economic conditions.

4. Results

4.1 Flood Extent Estimation

Table 1 Total flooded area

Parameter	Value
Total Flooded Area (m²)	377,833,188.33
Total Flooded Area (km²)	377.83

The flood extent analysis reveals that approximately **377.83 km²** of land in Sylhet was inundated during the 2017 flood event. The spatial distribution of flooding indicates that low-lying floodplain and haor regions were most severely affected.

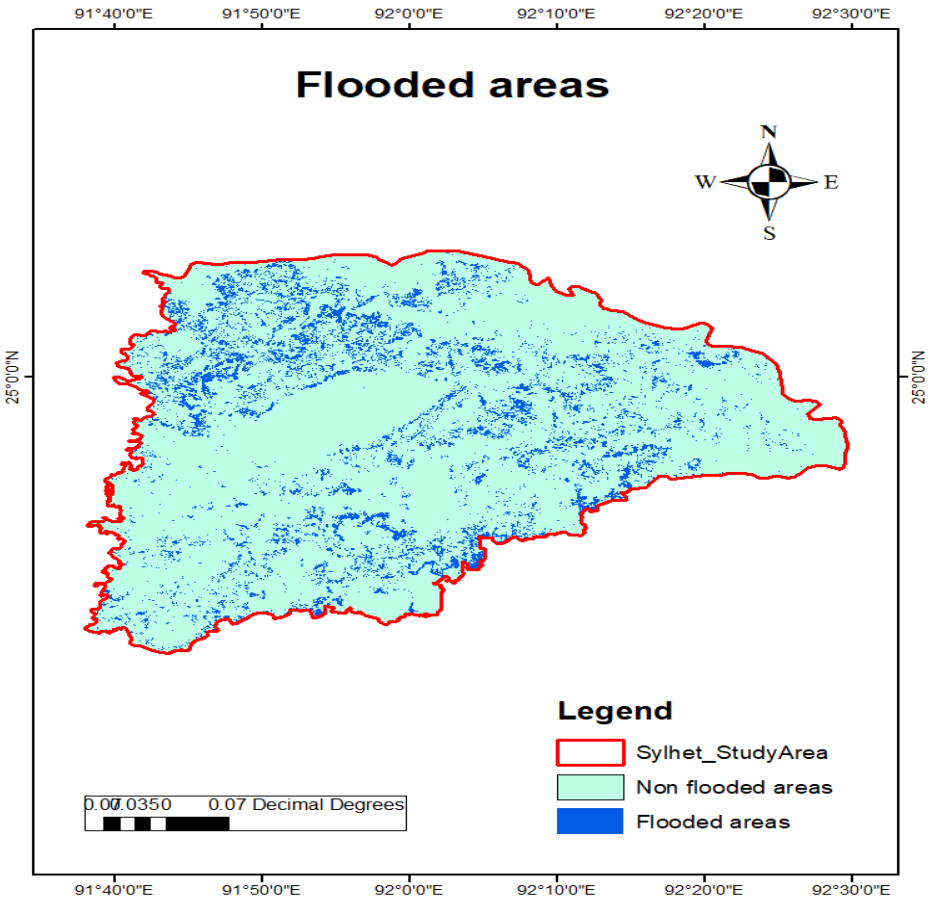

Figure 3 Flooded areas in Sylhet

4.2 Regression Dataset

Parameter	Value
Number of Samples	725
Variables Used	NTL Recovery, Population Density, RWI, Road Density

4.3 Regression Results

Variable	Coefficient (β)	Interpretation
Intercept (β_0)	1.7283	Baseline recovery level
Population Density (β_1)	-0.00295	Slight negative impact on recovery
Wealth Index (β_2)	-1.5242	Strong negative relationship
Road Density (β_3)	3.1544	Strong positive influence

The regression results provide an integrated understanding of how socio-economic factors influence post-flood recovery. Based on 725 observations, the model indicates that recovery, as measured by the NTL recovery index, varies significantly across the study area. Infrastructure, represented by road density, emerges as the most influential factor, showing a strong positive association with recovery. In contrast, population density and relative wealth index exhibit negative relationships with recovery levels.

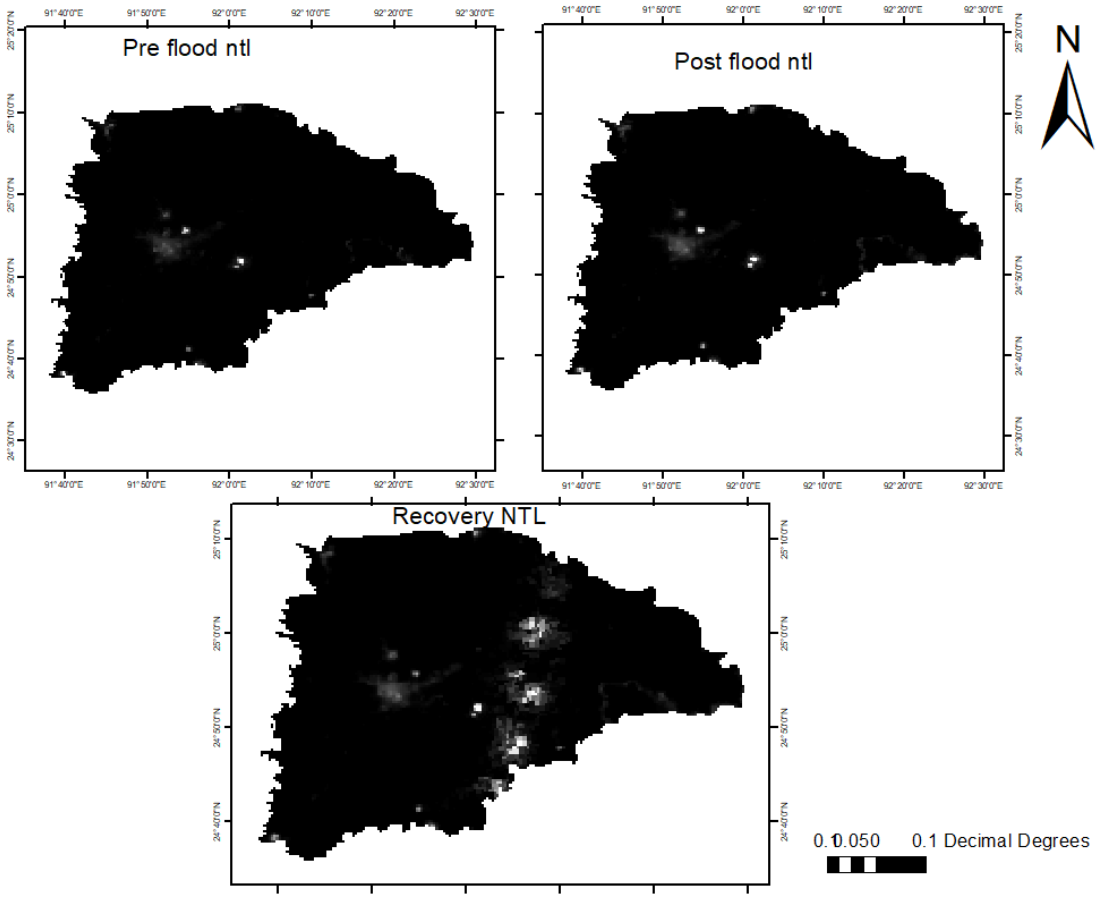

Figure 4 Nighttime light data of Pre flood, Post flood and Recovery

The findings suggest that areas with better accessibility and connectivity recover more efficiently, likely due to improved movement of resources and services. Conversely, densely populated areas may experience slower recovery due to increased pressure on infrastructure and resources. The negative relationship observed with the wealth index may reflect data limitations, spatial inconsistencies, or the inability of NTL data to fully capture economic dynamics, particularly in rural settings.

Overall, the regression model highlights that post-flood recovery is a complex and multi-dimensional process, with infrastructure playing a central role, while demographic and economic factors exhibit less predictable influences.

5. Discussion

The flood extent analysis confirms that the 2017 flood event had a substantial impact on the Sylhet region, inundating approximately **377.83 km²** of land. The application of Sentinel-1 SAR data proved highly effective for flood detection, primarily due to its ability to penetrate cloud cover and operate independently of weather conditions. This makes SAR-based approaches particularly suitable for flood monitoring in monsoon-affected regions like Bangladesh.

The regression analysis provides important insights into the spatial dynamics of post-flood recovery. The results indicate that infrastructure, represented by road density, plays a crucial role in facilitating recovery processes. Areas with higher road density likely benefit from improved accessibility, enabling faster distribution of relief, restoration of services, and resumption of economic activities.

In contrast, the observed negative relationship between the relative wealth index and recovery is somewhat unexpected. This counterintuitive finding may be attributed to several factors, including potential data limitations, spatial mismatches between datasets, or the nature of night-time light data itself. In economically active areas, disruptions caused by flooding may lead to a more pronounced decline in night-time light intensity, thereby reflecting slower apparent recovery. Additionally, NTL data may not adequately capture economic activities in rural or low-electrification areas, which can influence the overall interpretation.

Similarly, the slight negative association between population density and recovery suggests that densely populated areas may experience greater challenges during post-disaster recovery. Increased population pressure can strain available resources, infrastructure, and services, thereby slowing down the recovery process.

6. Conclusion

This study demonstrates the effectiveness of integrating remote sensing techniques with socio-economic data to assess flood impacts and recovery patterns. The use of Sentinel-1 SAR data enabled accurate detection of flood extent, revealing that approximately 377.83 km^2 of land in Sylhet was affected during the 2017 flood event. The incorporation of VIIRS night-time light data provided valuable insights into post-flood recovery dynamics, offering a proxy for economic and infrastructural restoration. The analysis further shows that infrastructure plays a pivotal role in enhancing recovery, as evidenced by the strong positive relationship between road density and recovery levels. In contrast, socio-economic variables such as population density and relative wealth index exhibit more complex and less predictable relationships with recovery, underscoring the multifaceted nature of disaster resilience. The study highlights the potential of geospatial approaches for disaster assessment and emphasizes the importance of infrastructure development in strengthening resilience to flood hazards.

7. Limitations

- Threshold-based flood detection may introduce classification errors
- GHSL data is an indirect proxy for road infrastructure
- Night-time light data may not fully capture rural economic activity
- Lack of ground truth validation
- Regression model does not include all possible influencing variables